

IABSE Congress Incheon 2026: Pre-Congress Workshop

Workshop title: Short Course on Structural Health Monitoring



Instructor(s): Eloi Figueiredo (Professor, Lusófona University), Ionut Moldovan, Michael Faber

Date: September 15, 2026 (in-person activities)

Venue: Songdo ConvensiA

Objectives: 1) Pose the SHM in the context of a statistical pattern recognition paradigm. 2) Understand the differences between quasi-static and dynamic monitoring. 3) Overview of sensors and DAQ hardware for designing an optimum instrumentation scheme for SHM. 4) Understand the applicability of finite element modeling and machine learning (unsupervised and supervised learning) for data enhancement and interpretation as well as for damage identification. 5) Understand the role of SHM to support climate change adaptation. 6) Pose the concept of probabilistic digital twins in the context of SHM. 7) Understand the role of SHM to support risk-informed integrity management. 8) Understand the goal of SHM with current limitations, grand challenges, and future trends.

Target audience: The course is tailored towards graduate students and/or practicing engineers working full-time in public and private institutions, such as authorities and contractors.

A brief description of the workshop: A 12-hour blended short course designed to pose the structural health monitoring (SHM) in the context of a statistical pattern recognition paradigm to support the damage identification process and risk-informed integrity management. The remote part introduces the concept of SHM and accelerates the learning process, while the in-person part focuses on bridging the gap between research and practical application. The techniques are shown with hands-on experiences applied to bridges. Unsupervised learning techniques as Gaussian mixture Models; supervised learning algorithms like artificial neural networks or support vector machines; and transfer learning methods as the transfer component analysis are all covered. The role of SHM to support the climate change adaptation of bridges is also discussed.